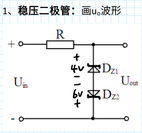
大题复习

2022年6月4日 18:06

1、**稳压二极管**：画u_o波形

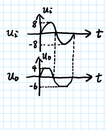


① $0<u_i<4V$ D_1止 D_2通 $u_o=u_i$

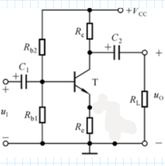
② $u_i\geq 4V$ D_1稳压D_1通 $u_o=u_{D1}=4V$

③ $-6V<u_i<0$ D_1通 D_2止 $u_o=u_i$

④ $u_i\leq -6V$ D_1通 D_2稳压 $u_o=u_{D2}=-6V$



2、**三极管共射**：画静态分析、动态分析、微变等效图；求I_E、I_B、I_C、U_{CE}；求A_u、R_i、R_o



静态分析：电容变断路；交流变短路（保留内阻）

$U_B=\frac{R_{b1}}{R_{b1}+R_{b2}}\cdot V_{CC}$

$U_E=R_e\cdot I_E$

$U_{BE}=U_B-U_E=U_B-R_e\cdot I_E$ U_{BE}已知

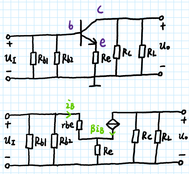
$I_E=\frac{U_B-U_{BE}}{R_e}$

$I_B=\frac{I_E}{1+\beta}$

$I_C=\beta I_B$

$U_{CE}=V_{CC}-I_C\cdot R_c-I_E\cdot R_e$

动态分析：直流变短路；电容变短路



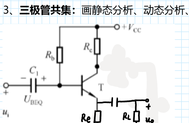
$u_i=i_B\cdot r_{be}+(1+\beta)i_B\cdot R_e$

$A_u=\frac{u_o}{u_i}=\frac{-\beta i_B(R_c//R_L)}{i_B[r_{be}+(1+\beta)R_e]}=\frac{-\beta(R_c//R_L)}{r_{be}+(1+\beta)R_e}$

$r_{be}=r_{bb'}+(1+\beta)\frac{26}{I_{EQ}(mA)}$（默认200Ω）

$R_i=R_{b1}//R_{b2}//[r_{be}+(1+\beta)R_e]$

$R_o=R_c$ 输出电阻看R_L



3、**三极管共集**：画静态分析、动态分析、微变等效图；求I_E、I_B、I_C、U_{CE}；求A_u、R_i、R_o

静态分析：

$U_E=R_e\cdot I_E$

$U_B=V_{CC}-R_b\cdot I_B$

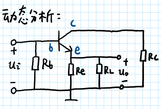
$U_{BE}=V_{CC}-R_b\cdot I_B-R_e\cdot I_E$

$I_C=\beta\cdot I_B$

$I_E=(1+\beta)I_B$

$U_{CE}=V_{CC}-I_C\cdot R_c-I_E\cdot R_e$

动态分析：

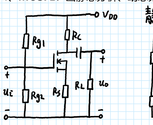


$A_u=\frac{u_o}{u_i}=\frac{(1+\beta)i_B\cdot(R_e//R_L)}{r_{be}\cdot i_B+(1+\beta)i_B\cdot(R_e//R_L)}$

$R_i=R_b//[r_{be}+(1+\beta)R_e]$

$R_o=R_e//(\frac{r_{be}+R_b}{1+\beta})$ ？

4、**MOSFE**：画静态分析、动态分析、微变等效图；求I_D、U_{GS}、U_{DS}、I_G；求A_u、R_i、R_o



静态分析：

$U_G=V_{DD}\cdot\frac{R_{g2}}{R_{g1}+R_{g2}}$

$U_S=I_D\cdot R_s$

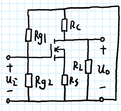
$U_{GS}=V_{DD}\cdot\frac{R_{g2}}{R_{g1}+R_{g2}}-I_D\cdot R_s$

$I_D=I_{DO}(\frac{U_{GS}}{U_{GS(th)}}-1)^2$

$U_{DS}=V_{DD}-R_c\cdot I_D-R_s\cdot I_D$

$I_G=0$

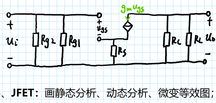
动态分析：



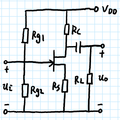
$A_u=\frac{u_o}{u_i}=\frac{-g_mu_{gs}\cdot(R_c//R_L)}{u_{gs}+g_mu_{gs}\cdot R_s}$

$R_i=R_{g1}//R_{g2}$

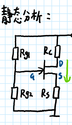
$R_o=R_c$



5、**JFET**：画静态分析、动态分析、微变等效图；求I_D、U_{GS}、U_{DS}、I_G；求A_u、R_i、R_o



静态分析：



$U_G=\frac{R_{g2}}{R_{g1}+R_{g2}}\cdot V_{DD}$

$U_S=I_D\cdot R_s$

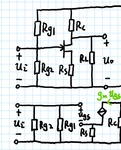
$U_{GS}=\frac{R_{g2}}{R_{g1}+R_{g2}}\cdot V_{DD}-I_D\cdot R_s$

$I_D=I_{DSS}(1-\frac{U_{GS}}{U_{GS(off)}})^2$

$U_{DS}=V_{DD}-I_D\cdot(R_c+R_s)$

$I_G=0$

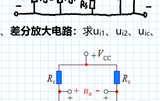
动态分析：



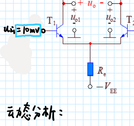
$A_u=\frac{-g_mu_{gs}\cdot(R_c//R_L)}{u_{gs}+g_mu_{gs}\cdot R_s}$

$R_i=R_{g1}//R_{g2}$

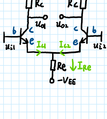
$R_o=R_c$



6、**差分放大电路**：求u_{i1}、u_{i2}、u_{ic}、u_{id}间的关系式；求I_C、U_C；求u_{id}、u_{ic}；求A_{ud}、A_{uc}、A_{ud1}、A_{uc1}；求u_o



静态分析：



$U_{i1}=U_{i2}=0$

$U_B=U_{B1}=0$

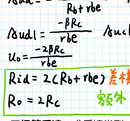
$U_{BE1}=U_{BE2}=-0.7V$

$I_{E1}=I_{E2}=\frac{1}{2}I_{RE}$

$I_{RE}=\frac{0-U_{BE1}-(-V_{EE})}{R_e}=\frac{0-U_{BE2}-(-V_{EE})}{R_e}$

$U_{CE1}=U_{CE2}=U_{C1}=U_{C2}=V_{CC}-I_{C1}\cdot R_{c1}-U_{E1}=V_{CC}-I_{C2}\cdot R_{c2}-U_{E2}$

动态分析：



$u_{i1}=u_{id}/2$ $u_{i2}=-u_{id}/2$ $u_{ic}=0$

$u_{id}=u_{i1}-u_{i2}$ $u_{ic}=\frac{u_{i1}+u_{i2}}{2}$

$A_{ud}=-\frac{\beta(R_c//\frac{R_L}{2})}{R_b+r_{be}}$ $A_{uc}=0$

$A_{ud1}=\frac{-\beta R_c}{r_{be}}$ $A_{uc1}=\frac{-\beta R_c}{r_{be}+2(1+\beta)R_e}$

$u_o=\frac{-2\beta R_c}{r_{be}}$

$R_{id}=2(R_b+r_{be})$ 差模输入电阻

$R_o=2R_c$ 双侧

差模

$u_{id}=u_{i1}-u_{i2}$

$u_{od}=u_{o1}-u_{o2}$

（放大倍数）$A_{ud}=\frac{u_{od}}{u_{id}}$

$A_{ud1}=\frac{-\beta R_c}{r_{be}}$

共模

$u_{ic}=\frac{u_{i1}+u_{i2}}{2}$

$u_{oc}=u_{o1}-u_{o2}$ （$u_{o1}=u_{o2}$）

$A_{uc}=\frac{u_{oc}}{u_{ic}}$（放大倍数）

$A_{uc1}=\frac{-\beta R_c}{r_{be}+2(1+\beta)R_e}$

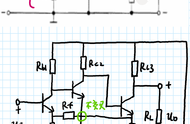
7、**三极管反馈**：求反馈类型；求反馈系数；求放大倍数

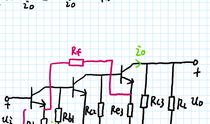
直接从U_o和反馈 ——电压反馈

反之 ——电流反馈

R_f连到U_i输入端 ——并联反馈

反之 ——串联反馈



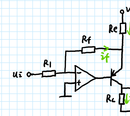


选择

$A_{uuf}=\frac{u_o}{u_i}=\frac{i_oR_c}{u_f}=\frac{i_o(R_{c3}//R_L)}{i_f\cdot R_{b2}}=\frac{R_{c3}//R_L}{R_{b2}}$

$F_{ui}=\frac{u_f}{i_o}=\frac{i_fR_{b2}}{i_o}=R_{b2}$

8、**综合反馈**：求反馈类型；求反馈系数；求放大倍数



选并负

$A_{uuf}=\frac{u_o}{u_i}=\frac{i_o\cdot R_c}{i_f\cdot R_1}=\frac{(R_f+R_e)\cdot R_c}{R_e\cdot R_c}$ $i_o=i_e+i_f$

$F_{ii}=\frac{i_f}{i_o}=\frac{R_e}{R_f+R_e}$ $\frac{i_f}{i_o}=\frac{R_e}{R_f+R_e}$